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SUB-SOIL INVESTIGATION REPORT FOR PROPOSED CONSTRUCTION OF 06 STORIED SERVICE BUILDING AT 250 BEDED DISTRICT HOSPITAL, SYLHET.

CLIENT
EXECUTIVE ENGINEER, PWD, SYLHET.

Verification Code: Xm6460M7
Verify report at: crtc-cee-sust.com

CONDUCTED BY
CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shah Jalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-45ft	09
02	Disturbed sample	01	0-45ft	09
03	Disturbed sample	01	0-45ft	09
04	Disturbed sample	01	0-45ft	09
05	-	-	-	-
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with borehole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf;$ $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 andTable11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	15	1.740	1.830	1.995	2.055	2.220	2.415	1.875
	5.0-10	17	1.964	2.066	2.252	2.320	2.506	2.726	2.116
BH02	2.5-5.0	6	0.742	0.781	0.851	0.877	0.947	1.030	0.800
	5.0-10	10	1.124	1.182	1.289	1.327	1.434	1.560	1.211
BH03	2.5-5.0	6	0.719	0.756	0.825	0.849	0.918	0.998	0.775
	5.0-10	8	0.936	0.984	1.073	1.105	1.194	1.299	1.009
BH04	2.5-5.0	6	0.650	0.683	0.745	0.767	0.829	0.902	0.700
	5.0-10	7	0.860	0.904	0.986	1.016	1.097	1.194	0.927
BH05	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH06	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH07	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH08	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH09	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH10	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.024	0.049	0.118	0.096	-	-	-	-	-	-
10	0.232	0.072	0.138	0.118	-	-	-	-	-	-
15	0.156	0.138	0.156	0.156	-	-	-	-	-	-
20	0.226	0.230	0.239	0.218	-	-	-	-	-	-
25	0.334	0.308	0.241	0.281	-	-	-	-	-	-
30	0.669	0.669	0.669	0.669	-	-	-	-	-	-
35	0.669	0.669	0.669	0.669	-	-	-	-	-	-
40	0.669	0.669	0.669	0.669	-	-	-	-	-	-
45	0.669	0.669	0.669	0.669	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	4.800	0.450	1.125	0.900	-	-	-	-	-	-
10	46.400	14.400	1.350	1.125	-	-	-	-	-	-
15	1.575	1.350	1.575	1.575	-	-	-	-	-	-
20	3.600	3.150	2.475	2.250	-	-	-	-	-	-
25	5.625	5.175	4.050	4.725	-	-	-	-	-	-
30	11.250	11.250	11.250	11.250	-	-	-	-	-	-
35	11.250	11.250	11.250	11.250	-	-	-	-	-	-
40	11.250	11.250	11.250	11.250	-	-	-	-	-	-
45	11.250	11.250	11.250	11.250	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.016	0.032	0.077	0.062	-	-	-	-	-	-
10	0.151	0.047	0.090	0.077	-	-	-	-	-	-
15	0.101	0.090	0.101	0.101	-	-	-	-	-	-
20	0.147	0.149	0.156	0.141	-	-	-	-	-	-
25	0.217	0.200	0.156	0.183	-	-	-	-	-	-
30	0.435	0.435	0.435	0.435	-	-	-	-	-	-
35	0.435	0.435	0.435	0.435	-	-	-	-	-	-
40	0.435	0.435	0.435	0.435	-	-	-	-	-	-
45	0.435	0.435	0.435	0.435	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	2.160	0.203	0.506	0.405	-	-	-	-	-	-
10	20.880	6.480	0.608	0.506	-	-	-	-	-	-
15	0.709	0.608	0.709	0.709	-	-	-	-	-	-
20	1.620	1.418	1.114	1.013	-	-	-	-	-	-
25	2.531	2.329	1.823	2.126	-	-	-	-	-	-
30	5.063	5.063	5.063	5.063	-	-	-	-	-	-
35	5.063	5.063	5.063	5.063	-	-	-	-	-	-
40	5.063	5.063	5.063	5.063	-	-	-	-	-	-
45	5.063	5.063	5.063	5.063	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

Based on the analysis and information provided the owner suggests the following recommendations for the proposed six-storied service building at 250 bedded district hospital.

Soil profile scenario

Overall soil information of the investigated site is very irregular in between the boreholes and vertical inconsistency exists in the soil formation. Reddish soft clay soil having very low bearing capacity exists in the top depth for all boreholes followed by a reddish stiff clay layer has been observed.

From the above discussion, it is very clear that the whole scenario of the proposed site is quite variable in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since very low bearing capacity is found in the topsoil layer, it is difficult to improvise any shallow foundation for the proposed site.

Moreover, a deep foundation (Bored pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found in the attached table (10 and 11) in the report, which is based on existing soil conditions.

Safety provisions

Since the proposed site is composed of soft cohesive and granular soil layers, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering the building load while designing either a shallow or deep foundation. Obviously, a **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of a shallow or deep foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix.

Special notes

These suggestions are tentative and the consultant or design engineer would decide the selection of type, depth, and dimensions of the foundation. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **45ft** as the client provided.

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